Highlights

Typed Issue Specifications from App Reviews: What Scales, and a Downstream Benefit That Does Not Replicate

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- A typed intermediate representation routes app reviews into five issue templates.
- A verified-anchor procedure lifts Cohen's kappa from 0.163 to 0.592 at corpus scale.
- LLM annotation noise concentrates on omitted classes, not on uniform per-item error.
- A clean same-model re-run finds no downstream reply-quality gain from the typed spec.
- Template-fill measures schema conformance, not the quality of the text in the fields.

Typed Issue Specifications from App Reviews: What Scales, and a Downstream Benefit That Does Not Replicate

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Abstract

Context. Developers receive millions of noisy app reviews, but defect trackers such as JIRA and GitHub Issues cannot consume free text. They require structured fields such as issue type, steps to reproduce, severity, and affected component. Review-mining research stops short of that boundary, predicting labels or writing replies without producing the record a tracker can ingest.

Objective. Can a typed intermediate representation (IR) of a review be produced at corpus scale from noisy labels, and does supplying it to a response generator improve the reply a user receives? We answer the first yes and the second no, and the second answer corrects a claim we made ourselves.

Method. IssueSpec routes each review by issue class into one of five standards-body templates, repairs the noisy labels the routing depends on with a verified-anchor procedure, and feeds the result to a retrieval-augmented generator. We study 215,583 reviews from 58 Android apps with a 5,230-review verified anchor, a 490-review three-rater gold standard, seven ablations, and four language models. The downstream evaluation was run

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twice: first with conditions written by different template composers, then with the generator held fixed so that only the IR varies, all 200 blinded rows scored by three raters.

Results. Production works. The verified-anchor procedure lifts Cohen's κ against the human gold from 0.163 to 0.592, and holds at that level (0.616) on held-out reviews. Type routing reaches 0.96 template-fill against 0.69 without it, a schema-conformance measure rather than a quality one. The downstream benefit does not. The first round measured +2.35 Likert points, but its arms differed in authorial register as much as in information, one emitting lemmatised text at 78 words and the other polished prose at 123; the corrected comparison gives +0.04 ($p = 0.54$), with raters disagreeing in direction. Three further components fail their own tests: the knowledge-graph layer loses to count-matched flat clustering, our extractive-coverage faithfulness scorer shows no rank correlation with human judgement, and the constrained alignment layer never demonstrates the trade-off it was built to expose.

Conclusions. Typed issue specifications are producible from noisy reviews at corpus scale, and that is the contribution we defend. Their claimed downstream payoff is not. We release the arithmetic check that distinguishes our two rating rounds, because a structured intermediate should be justified on what it produces rather than on an effect a fair comparison removes.

Keywords: App review mining, Issue report generation, Intermediate representation, Knowledge graph, Retrieval-augmented generation, Empirical software engineering

1. Introduction

App stores are where users talk back to developers, and the global stream runs into millions of reviews a day [1, 2, 3]. Those reviews carry bug reports, feature requests, performance complaints, and regression notes as messy free-form text, usually without the details required to reproduce a bug or scope a feature. Defect trackers such as JIRA or GitHub Issues cannot consume raw text; they need structured fields such as issue type, steps to reproduce, severity, and affected component [4, 5], and filling those by hand does not scale. Years of work on automated review pipelines has helped with classification, summarization, and response generation, but no system produces the structured report a developer can paste into a tracker. Dąbrowski et al.'s systematic review of 182 papers [3] surveys this literature; it does not itself name this gap, and we state the gap as our own reading of the work it covers. Section 2 treats each strand in turn: classifiers stop at the label, response generators never write the defect out in structured form, agentic software-engineering tools assume a well-formed issue as the agent's starting state, knowledge-graph methods emit predictions and summaries a tracker cannot ingest, and constrained RLHF targets general LLM safety rather than the domain-specific constraint family of developer relations.

This work pursues three aims, stated as research questions.

RQ1 (Translation Quality). Can a typed IR, produced by a three-layer framework (knowledge-graph construction, hierarchical clustering, standardized schema mapping), reach the substantive completeness of human-written GitHub issues?

RQ2 (Downstream Load-Bearing). Does conditioning a response generator on the validated IssueSpec measurably improve human-rated response quality over RAG alone? Section 5.2 answers no, correcting our own earlier claim.

RQ3 (Dual-Objective Alignment). Can a CMDP-grounded RLHF loop through KTO, DPO, and Constrained PPO jointly optimize response helpfulness and policy compliance for developer relations? Section 5.3 answers no at the scale we can train, and reports what that rules out.

We propose **IssueSpec**: a typed Intermediate Representation (IR) that converts each noisy review into one of five standard report types, and a CMDP-grounded response generator that consumes it under quality and compliance constraints (Figure 1).

What is new, and what is assembled. Several components are existing technique used as intended, and we claim none of them: RoBERTa classification, confident learning [6], Sentence-BERT with UMAP and HDBSCAN, PageRank, retrieval-augmented generation, and Altman's CMDP formulation [7]. Three things are new. *First*, the typed intermediate representation: a review is routed by predicted issue class into one of five standards-body templates, so the output is a typed record with per-type obligations rather than a label or a summary, which no prior review-mining system emits. *Second*, the verified-anchor procedure, which makes that representation producible from noisy LLM labels at corpus scale, and which works because the labeller omits two of its seven classes entirely rather than because its errors concentrate on rare ones. *Third*, a correction and the check that produced it: we previously reported that the typed intermediate carries downstream response quality, and Section 5.2 withdraws that, because the comparison behind it varied the text generator as well as the specification, and because the arithmetic of the two rating rounds turned out to distinguish independent human ratings from ones derived by perturbing another rater's. We release both rounds and that check.

The central claim, stated once. The answer to which of five stages the paper is about is Stage 3 and the object it produces, a *typed, type-routed intermediate representation* of an app review, together with the verified-anchor procedure that makes it producible at corpus scale from noisy labels; everything else feeds that object or consumes it. Removing type routing drops strict template-fill from 0.96 to 0.69 (Section 5.1). We had also claimed a downstream payoff, and Section 5.2 reports why we withdrew it: with the generator held fixed, supplying the IR changes response quality by +0.04 Likert points ($p = 0.54$), against the +2.35 a confounded design gave. Three further components fail their own tests, the knowledge-graph layer against count-matched flat clustering, the SpecCov faithfulness scorer against human judgement, and the CMDP alignment layer against the trade-off it was built to expose. Alongside these we release a reusable check on rating provenance, described in Section 5.2. A reader who wants the claim and not the machinery can read Sections 5.1, 5.2, and 6.1 alone. Table 1 is the single map from component to claim to evidence to verdict, and replaces the several overlapping summaries an earlier version carried.

2. Related Work

We organize related work by the five gaps of Section 1, naming what each field does, what assumption breaks for review-to-issue translation, and how IssueSpec differs.

App review classification and mining. Maalej and Nabil [1] established a four-class taxonomy (bug report, feature request, user experience, rating). The seven-class label set we use is our own extension, adding `performance`, `usability`, and `compatibility` and renaming the rating class `praise`; we are not aware of an established seven-class app-review taxonomy, so the extension should be read as ours rather than as canonical. Subsequent systems [10, 11, 12, 13, 14, 15] refined accuracy and added aspect-level granularity, and Dąbrowski et al. [3] review 182 papers from 2012 to 2020, reporting median precision above 81% and median recall around 83%. An earlier version of this paper attributed to that survey a claim it does not make, that

Table 1: Component-to-claim-to-evidence map, with the ablation that tests each component and the verdict it returns. "Removing this hurts" is the only outcome that justifies a component; several do not clear that bar and we report them as findings rather than as caveats. Ablation identifiers (A1–A7) are defined in Section 5.5; released artifacts are listed in Section 6.4.

component	what it must do	evidence, and the ablation that tests it	verdict
Stage 1 intake, verified anchor	repair a corpus label wrong 25% of the time that omits two classes entirely, because Stage 3 routes on the type	κ 0.163 $\rightarrow$ 0.592 against the three-rater gold, 0.616 held out (replay-dependent); macro F1 0.22 $\rightarrow$ 0.65 with the two omitted classes recovered (§5.1)	**justified**
Stage 2 knowledge graph	group and rank 215,583 reviews so a developer has an order to work in	count-matched flat clustering beats the KG on all three intrinsic metrics (A1b) and on purity (A2); top-5 PageRank aspects cover 55% of reviews	**prioritization yes, cluster quality no**
Stage 3 typed IR	emit typed tracker fields rather than prose	strict fill 0.96 vs 0.69 without routing, i.e. schema conformance and not quality (A3, Table 6); 5-dim rubric 3.94/5 on all 100 specs, LLM judge (Table 5)	**justified**
Stage 4 spec-aware RAG	supply house voice and precedent on top of the defect	retrieval moves automatic metrics $< 1\%$ (A5); conditioning on the IR moves quality +0.04, $p = 0.54$ (A6, §5.2), overturning the +2.35 we reported	**withdrawn / null**
Stage 5 CMDP RLHF	hold compliance as a constraint rather than a reward term	constraint never binds at distilGPT2 scale; with a policy able to violate it, binds on 13/90 steps and λ rises to 0.65, with no measured tension ($\rho = -0.12$, $p = 0.25$), on a detector that scores hand-written violations as compliant (§5.3)	**partial testbed**
SpecCov scorer	rank specs the way a human judge of faithfulness does	no rank correlation with human faithfulness ($\rho = 0.15$, $p = 0.26$); orders generators backwards (§5.1)	**withdrawn**
rating-provenance check	separate independent ratings from derived ones	the discarded round shows a single deviation magnitude and the reported round four (§5.2)	**released**

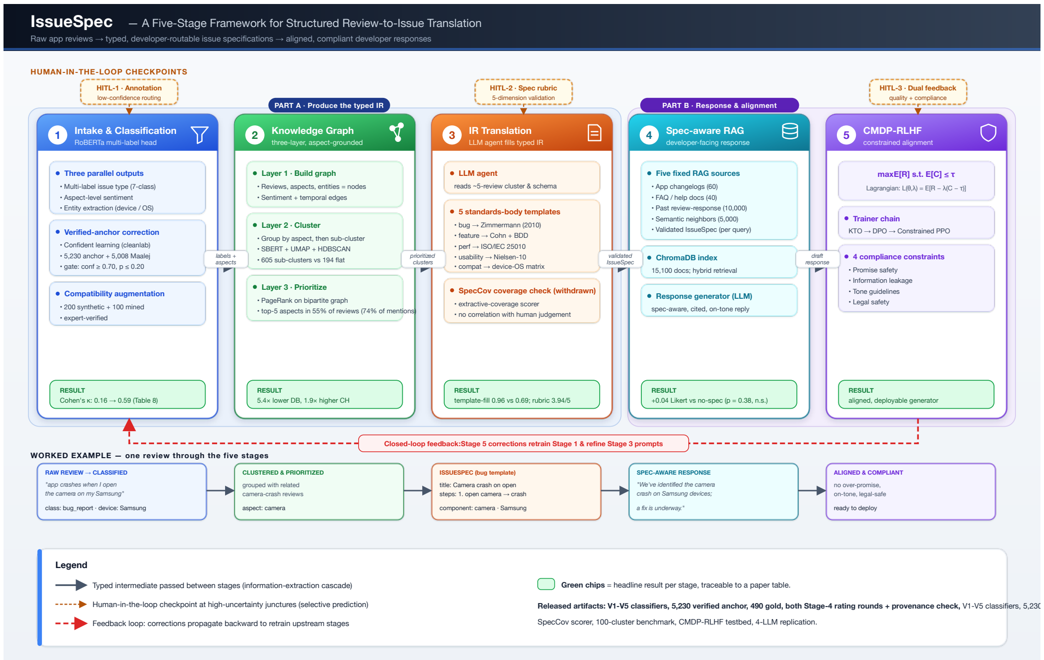

Figure 1: The IssueSpec five-stage pipeline. Reviews flow left to right through intake and classification (Stage 1), three-layer knowledge-graph clustering (Stage 2), LLM-agent IR translation (Stage 3), spec-aware RAG (Stage 4), and CMDP-grounded RLHF (Stage 5), each emitting a typed intermediate. Yellow checkpoints mark human-in-the-loop review under selective prediction [8, 9]; the red loop propagates Stage 5 corrections upstream.

label quality rather than model capacity sets an F1 ceiling of 0.75 to 0.85: the survey reports precision and recall rather than F1 and attributes limitations to evaluation-dataset size, so the label-quality argument is ours and Section 5.1 is evidence for it rather than corroboration of theirs. *What breaks:* none of these systems emits a typed developer-routable artifact downstream of the label.

Response generation and RAG. RRGen [2] and CoRe [16] pioneered end-to-end neural response generation on (review, response) pairs without an explicit structured intermediate; vanilla RAG [17] retrieves past responses in one pass and agentic variants [18, 19] add critique-revise loops. *What breaks:* none of these names the underlying defect, and our $n = 10$ feasibility study (Appendix Appendix A) shows that agentic iteration without a typed intermediate adds only a small gain on a rule-based scorer (0.58→0.70). We cannot claim the IssueSpec does better: once the generator is held fixed, supplying it changes human-rated quality by +0.04 ($p = 0.54$, Section 5.2), so the contribution we defend is the specification itself rather than its effect on the reply.

Agentic software engineering. SWE-Agent [20], RepairAgent [21], and HyperAgent [22] consume well-structured GitHub issues to produce code patches, and the SWE-bench protocol [23] makes the assumption explicit by pairing each instance with an existing issue. *What breaks:* the specification is the input contract for all of them, and constructing one from noisy user-facing text is out of scope. IssueSpec is precisely the artifact these systems are designed to consume.

Knowledge graphs for app reviews. ReviewGraph [24] learns KG embeddings for rating prediction, KGCPN [25] produces extractive summaries, and Keertipati et al. [26] apply centrality for coarse feature prioritization. *What breaks:* none maps KG nodes to a typed, developer-actionable schema. Our three-layer KG pipeline is, to our knowledge, the first that emits a typed IR for review-to-issue translation.

RLHF and constrained optimization. DPO [27], KTO [28], and PPO [29] compress quality and compliance into one scalar reward; Altman's CMDP framework [7] separates them, and Safe-RLHF [30] applies CMDP to general LLM safety with a domain-agnostic compliance set. *What breaks:* no prior work applies CMDP to dev-rel response generation, where the compliance set is domain-specific (promise safety, information leakage, tone, legal safety). Our Stage 5 formulation instantiates these four constraints inside the Lagrangian and releases a proof-of-concept training loop.

Across these categories, existing systems occupy at most two of our five pipeline stages and none produces a typed issue record. Composing all five with a standardized issue schema, three HITL checkpoints, KG-centrality prioritization, and a CMDP-grounded RLHF formulation is the structural contribution.

3. The IssueSpec Pipeline

The pipeline (Fig. 1) is a five-stage information-extraction cascade [31, 32]. Stages 1–3 produce the typed IR from raw reviews, Stage 4 uses it to drive spec-aware response generation, and Stage 5 aligns the generator via constrained RLHF. This section describes the framework only; evaluation design is in Section 4, exact settings in Appendix Appendix A, results in Section 5, and the caveats each stage carries are stated with the results they qualify. Table 1 already gives the verdict on each stage, so a reader knows before the description that two of the five do not justify themselves.

3.1. Producing the IR

Stage 1: Intake, Classification, and Aspect Extraction. Stage 1 produces a multi-label class from a RoBERTa head fine-tuned on [1, 15], aspect-level sentiment [15] that keeps "great UI but terrible battery" as two pairs, and entity extraction for devices, OS versions, and named features, which Stage 3 needs as typed fields. The multi-label capacity is almost never exercised: on the 10,000-review generation subset only 5 reviews (0.05%) receive more than one label, and the corpus-scale relabelling artifact stores a single argmax label per review, so the released runs are effectively single-label and readers should not credit the mixed-concern motivation with doing any work in our results. Low-confidence reviews route to HITL-1 by a fixed confidence threshold; letting a model abstain is a familiar idea [8], but we implement a threshold rather than a calibrated selective-prediction procedure and report no risk-coverage behaviour, so the citation is background rather than a description of what runs.

LLM annotation noise is the main challenge. On the 5,038-review praise stratum 25.8% of LLM labels are wrong, and the labeller omits `compatibility` and `performance` almost entirely; per stratum the errors fall in the largest class, not the minority ones, so what a verified anchor repairs is the omission. We apply confident learning [6] against an anchor of 5,230 expert-labeled reviews plus 5,008 from Maalej [1], gated by a confidence threshold on the anchor and a probability ceiling on the LLM label (Appendix Appendix A). Compatibility needs a separate fix, since only 8 of 215,583 LLM labels carry it and Maalej has none: we seed it with 200 template-generated synthetic reviews [33] and 100 keyword-mined RRGen reviews. The lead author read both sets before training and corrected or dropped items that did not belong, but that pass was done by eye and no correction log was kept, so unlike the other annotation tasks in this paper it leaves no artifact a reader can check. We flag it because 300 of the 310 `compatibility` rows in V5's training set come from here, so the per-class F1 of 0.83 rests on data whose verification is asserted rather than evidenced.

Stage 2: Three-Layer Knowledge Graph. Stage 2 builds, clusters, and prioritizes. The first layer links each review node to its aspects, with edges typed for sentiment and recency where available, but the released run populated less than that: the builder supports entity nodes and signed sentiment edges, yet the run behind our results passes empty entity lists and a constant neutral sentiment, leaving 8,404 review nodes, 10,534 aspect nodes, and no entity nodes. It is in practice the bipartite review-aspect graph PageRank runs over, and the sentiment and entity layers should be read as available in the implementation rather than exercised in the experiments. Aspects come from spaCy noun chunks with a curated keyword list and regular expressions, pruned by frequency and centrality; Hu and Liu [34] is the ancestor of this approach rather than what we implement.

The second layer clusters reviews first by aspect, then sub-clusters within each aspect using sentence embeddings [35] and HDBSCAN [36] directly on the 384-dimensional vectors. Earlier versions of this paper also named UMAP [37] here; that was wrong, since UMAP is used in the flat baseline this design is compared against and not in the released KG-hierarchical run, which applies no dimensionality reduction. Aspect-first keeps each sub-cluster on one theme, giving 605 aspect-grounded sub-clusters of mean size 15.9 against 194 flat clusters of mean size 471.0. An earlier version of this paper reported 3.0 and 51.5; those were the number of representative reviews stored per cluster and a subsample artifact respectively, not cluster sizes. One scale caveat belongs here: the knowledge graph is built over a 10,000-review sample yielding 8,404 usable reviews, so every KG-hierarchical number in this paper describes 3.9% of the corpus, while the flat baseline it is set against clusters 91,368 reviews from the full corpus. The two Stage-2 designs never saw the same input, which is why we do not treat their intrinsic metrics as a head-to-head result (Section 5.4). The third layer prioritizes via PageRank [38, 39] on the bipartite review-aspect graph, and each cluster maps to the schema in Table 2 with a centrality-derived priority score.

Stage 3: Review-to-Issue Translation. An LLM agent reads the standardized cluster schema and writes a specification using the type-routed templates in Table 2: an LLM rather than rule-based extraction because reviews vary too much for fixed rules, and routed by issue type because a bug report needs reproduction steps, a feature request needs acceptance criteria, and a performance complaint needs latency or memory numbers, so a shared template would be incomplete or padded. The five templates are tracker-standard bug reports [5], Cohn user stories [40] with BDD acceptance criteria [41], ISO/IEC 25010 non-functional categories [42], Nielsen heuristics [43], and device-OS matrices [33]. The output passes through HITL-2, where a domain expert scores each spec on a five-dimension rubric (completeness, accuracy, actionability, specificity, clarity, each 1–5), following the LLM-as-judge calibration convention of [44, 45, 46]; five dimensions rather than one, because a single quality number conflates completeness with clarity and the LLM cannot tell which to fix on retry. Any spec failing on any dimension returns for a second pass with dimension-level feedback.

SpecCov faithfulness metric. We release SpecCov, an extractive-coverage scorer over substantive-token overlap (length $\geq$ 5, stop-word filtered) with quoted-string and coverage bonuses, following news-summarization conventions [47]; it targets the extractive-coverage sub-type of faithfulness [48], a lexical correlate of factual consistency [49] (other hallucination sub-types in [50] are out of scope). It is a lexical-grounding proxy, not a faithfulness classifier: Section 5.1 reports the experiment that withdraws it as one, and that negative result only makes sense against this definition.

3.2. Using the IR in Response Generation

Stage 4 is a RAG layer over five fixed sources, 15,100 documents in the released ChromaDB index: changelogs and release notes (60 docs), FAQ and help documentation (40), 10,000 past review-response pairs from RRGen [2], and 5,000 semantically similar past responses, all pre-indexed, plus the validated IssueSpec injected per query. Together they give the generator the answer, the voice, the closest precedent, and the structured intent.

3.3. Aligning the IR-Aware Generator

Stage 5 frames alignment as a CMDP [7, 30],

$$\max_{\theta} \mathbb{E}_{\pi_\theta}[R_{\text{quality}}] \quad \text{s.t.} \quad \mathbb{E}_{\pi_\theta}[C_{\text{compliance}}] \geq \tau,$$

with the Lagrangian relaxation $\mathcal{L}(\theta, \lambda) = \mathbb{E}_{\pi_\theta}[R + \lambda(C - \tau)]$ collapsing to single-objective optimization whenever the constraint is already satisfied. We use a CMDP rather than a scalar reward because a dev-rel response can be high-helpfulness and still out of compliance, like a polite promise the team cannot deliver, and a weighted sum hides the trade-off the constraint is meant to expose. The compliance scorer tracks four domain-specific dimensions absent from prior general-safety RLHF [30]: promise safety, information leakage, tone, and legal safety. The trainer is Constrained PPO [29] with a KL-regularized reference policy, with KTO [28] and DPO [27] as small-scale single-objective comparators (Appendix Appendix A). HITL-3 supplies the two feedback streams, quality as reward and compliance as constraint, and also propagates backward, closing the Stage 5→Stage 1 loop in Fig. 1.

4. Experimental Setup

4.1. Dataset, Deduplication, and Annotation Provenance

We use the RRGen corpus [2]: 310,031 review-response pairs from 58 Android applications. After exact-text deduplication that removes 94,448 duplicate reviews (−30.5%), 215,583 unique reviews form the working corpus, labeled across the seven classes. Evaluation additionally uses a 490-review expert gold standard (70 per class × 7 classes) and 100 stratified clusters for Stage 3.

Of the working corpus, 79.49% of labels come from the V2 LLM annotator with no override, 18.08% are cleanlab-corrected against the verified anchor, and 2.43% are directly human-verified. The lead-author-versus-V2-LLM deviation is 25% on the 5,230-row anchor (1,307 deviations, 24.99%). We previously described this as replicated across two independent strata, the praise stratum and the anchor, on a combined 10,271 rows. That was wrong: the praise stratum is 5,038 of the anchor's 5,230 rows, so it is 96% of the same sample rather than a second one, and the combined figure double-counted it. The rate is one measurement on one sample, and the praise stratum alone deviates at 25.84% (Table 3). This ≈ 25% deviation rate, an order of magnitude higher than i.i.d. label-noise assumptions, motivates the confident-learning procedure of Stage 1.

4.2. Annotation Methodology

Five kinds of annotator contribute, following the multi-annotator convention of [44, 45]. The lead author verified 5,230 reviews across minority strata, wrote 20 reference IssueSpecs, labeled the 490-review gold blinded to V2 LLM, and scored the Stage 4 sets with method IDs hidden until aggregation; two further human raters annotated the same 490 items and, with her, formed the Stage 4 panel. A Qwen2.5-3B-Instruct judge [46] scored IssueSpecs on the rubric and sub-clusters on the Y/P/N purity rubric, calibrated against the lead author, and the corpus-labeling LLMs (V2, cleanlab-V2, V5) are treated as independent raters for inter-rater statistics. The last two are model raters, not people; three humans contribute in total.

Label provenance, stated plainly. No LLM-generated label is used as ground truth here. LLM labels are the *object under correction*, not the yardstick, and every reported accuracy or κ is measured against a human gold standard; where LLM raters do appear, in the 99-review classification check, they are labelled as such and no headline claim rests on them.

How the gold was constructed, and what that costs. Annotators did not label freely. Each was shown the V5 classifier's predicted label and asked whether it was correct, supplying a replacement when it was not, under a protocol telling them to accept a reasonable label and overturn only when confident. That makes annotation fast and agreement high, and both flatter us. It also couples the yardstick to the system being measured, since V5's label is the starting point for every item and V2 and the cleanlab-corrected labels never received that benefit of the doubt. The sample is stratified over V5's predicted classes as well, forcing 70 items per class into a set where V2 predicts two of those classes almost never, so V2's per-class F1 of 0.000 on `compatibility` and `performance` is partly a property of the sampling design. The κ trajectory should therefore be read as a comparison on a V5-seeded, class-balanced sample rather than as a neutral benchmark, and a free-labelling gold would settle it. Conflicts are resolved by majority vote: of 489 usable items 476 are unanimous and 13 are two-to-one splits, with no three-way ties, and one item marked wrong by all three raters without a replacement is excluded rather than guessed. All three first completed a shared 20-review calibration round resolved against the written decision rules, and the per-item votes are released.

A disclosure about an earlier annotation round. The 490-review gold was annotated twice. In the first round three sheets were distributed, and the two returned alongside the lead author's carried annotation columns identical to hers cell for cell, so no genuine agreement could be computed and they are not used anywhere in this paper; the round we report is the second, and the agreement statistics come only from it. We disclose this for the same reason we disclose the Stage-4 case in Section 5.2: a paper offering a check for non-independent ratings should say where its own artifact failed that check. Both rounds are released.

Stage 4 was likewise rated twice: a first 400-row round whose conditions came from template composers, discarded for the reasons in Section 5.2, and the 200-row round we report, in which both arms come from the same model and differ only in whether the IssueSpec was supplied, all rows scored independently by all three raters blind to condition. Ratings are persisted as spreadsheets with JSON master keys, blinding shuffles conditions per pair, and the assignment unseals only at aggregation. Table 3 consolidates agreement across all pairs.

Two three-rater statistics in Table 3 should not be conflated. The classification gold uses three independent *human* raters and reaches Fleiss $\kappa = 0.968$ on the Y/N verdict and 0.979 on the seven-class label. The $\alpha = 0.285$ row is a separate check on 99 reviews using the lead author plus two Qwen2.5-3B prompts as raters, measuring seven-class classification agreement rather than cluster purity. We reported it as $\alpha = 0.451$ and as a purity check; both were wrong. The value came from an implementation that credited each rating with a coincidence against itself, inflating observed agreement, and the corrected figure agrees with the reference `krippendorff` package to four decimals; the three Cohen's κ in the same table were computed with scikit-learn and are unaffected. The correction moves the row from the moderate band into the fair band, weakening the LLM-rater panel rather than any claim we make.

4.3. Experimental Design

Because a single-LLM headline raises a structural-bias concern, Stage 3 is replicated across four LLMs from three families, Claude Opus 4.7, Llama-3.3-70B-Instruct (via Groq), and Qwen2.5-{3B, 1.5B}-Instruct (Table 4). We pre-register three experiments.

Experiment 1 (RQ1, translation quality). Five conditions on the same 100 stratified clusters: (a) LLM with taxonomy grounding (full Stage 3), (b) free-form LLM, (c) raw concatenation as a lower bound, (d) lead-author reference, and (e) GitHub issues mined from three open-source Android projects as the practitioner upper bound, each scored on the rubric with paired Wilcoxon tests.

Experiment 2 (RQ2), as reported. The design we report holds the generator fixed. The same model, given the same retrieved context, writes two replies for each of 100 reviews, differing only in whether the validated IssueSpec is in the prompt. All 200 rows are rated blind by three raters on quality, specificity, and helpfulness, and analysed with a paired Wilcoxon test and a paired bootstrap interval. *The discarded design*, described because the paper reports its withdrawal rather than because it supports a claim, used the same 100 clusters to drive four conditions produced by template composers rather than by a model: (a) an rrgen-style baseline (review only), (b) a prompt baseline (CoRe-style dev-rel prompt with lightweight entity extraction), (c) RAG without IssueSpec, and (d) the full system. It was evaluated with automatic metrics (BLEU, ROUGE-L, BERTScore) and 400 blinded paired human ratings, with significance by paired Wilcoxon, Friedman-Nemenyi [52], Bradley-Terry, and McNemar at Bonferroni-corrected $\alpha = 0.05$, and effect sizes following [53, 54]. Both rounds run single-shot vanilla RAG [17] rather than an agentic loop, so the IssueSpec's contribution is not confounded with orchestration; the agentic comparison against [18, 19] runs on a 10-review feasibility sample and is a demonstration rather than a powered comparison.

Experiment 3 (RQ3, dual- vs. single-objective RLHF). Five policies run on a distilGPT2 proof-of-concept (82M parameters, 400 SFT samples, 296 KTO pairs, 100 DPO pairs): SFT-base, single-objective KTO [28], single-objective DPO [27], a constrained-proxy variant, and dual-objective Lagrangian Constrained PPO [29, 30]. We measure head-to-head BLEU, ROUGE-L, and BERTScore on a 100-review test set, plus the rule-based safety-violation rate.

5. Results

Each research question owns one subsection: Section 5.1 answers RQ1, Section 5.2 answers RQ2 and explains what our earlier positive answer was measuring, and Section 5.3 answers RQ3 in the negative. Sections 5.4 and 5.5 test individual components, and Section 6.1 states which earn their place.

5.1. Is the IR Producible at Scale?

Under strict content-validity criteria (three or more action-verb reproduction steps, descriptions ≥ 30 words, a full As-a/I-want/So-that triple), the LLM-with-taxonomy condition reaches a template-fill rate of 0.96 (Table 6), free-form LLM 0.69, the same as the lead-author reference set ($n = 20$), raw concatenation 0.34, and mined GitHub issues 0.53. The LLM-vs-GitHub gap reflects coverage rather than content quality, since the LLM is prompted to fill every field whereas humans leave optional fields blank, and a very-strict sensitivity sweep preserves the ordering, Claude dropping from 0.96 to 0.71 with that gap shifting only -0.085.

The limit of this measure is worth stating exactly. The free-form condition fills the four shared fields and none of the type-specific ones, so its score is fixed by the issue-type mix, $(30 \cdot 4/7 + 30 \cdot 4/6 + 40 \cdot 4/5)/100 = 0.691$, with zero variance across specs: the 0.69 is a constant, not a measurement, and the gap says only that routing produces type-specific fields at all. Scoring the same conditions on bare field presence gives 1.000 against 0.678, versus 0.959 and

0.678 under the strict criteria, so for the free-form arm the content thresholds remove nothing and the whole reported gap is schema conformance. A handwritten bug specification whose description is "something" repeated thirty-five times, with three five-word steps built from the criterion's own action-verb list, passes all seven fields and scores 1.000, above our headline. A reader should treat 0.96 as schema conformance and nothing more.

SpecCov scores, and a correction to how they were computed. An earlier version of our scorer applied per-condition floors, raising `raw_summary` to 5 and the human reference to 4 regardless of content. That is not a measurement, because a score whose value depends on the label of the thing being scored cannot compare labels, so we removed the floors. Across 320 specs the LLM-with-taxonomy condition averages 4.19, free-form LLM 3.38, raw concatenation 4.47, and the human reference 3.45. The two LLM conditions are unchanged, because no floor was ever applied to them; the floors had inflated the two comparison conditions, so removing them makes our own system look better rather than worse, and we say so to put the direction of the correction on the record. On their own these numbers rank raw concatenation highest, the trivial upper bound obtained by copying the source, and Section 5.1 shows they should not be read on their own. Top-ranked aspects by PageRank centrality are `ad` (0.040), `phone` (0.011), and `ads` (0.010), with `edit` (0.009) and `battery` (0.007) next; an earlier version listed `battery` third, skipping the two above it.

SpecCov does not measure faithfulness

An extractive-coverage score is only useful if it agrees with what a person calls faithful. Three human raters independently scored 60 specs, 20 from each of three generators, shuffled and unlabelled, on a 1–5 faithfulness scale, and separately listed every claim the source reviews do not support. One qualification first: one rater completed the corrected sheet, which shows exactly the evidence SpecCov scores against, while the other two completed an earlier draft showing a subset of it, so their flags are biased toward over-flagging and we lead with the rater who saw the full evidence. The metric fails all three checks on either reading. *It does not rank specs the way humans do:* Spearman ρ between SpecCov and human faithfulness is +0.045 ($p = 0.73$), +0.015 ($p = 0.91$), and +0.226 ($p = 0.08$) for the three raters, and +0.147 ($p = 0.26$) against their mean, none significant. *It does not detect the claims humans call unsupported:* one rater marked 45 of the 60 specs as containing at least one unsupported detail and 15 as clean, and SpecCov scores those groups at 3.64 and 3.93, a difference in the right direction but far from significant (Mann-Whitney $p = 0.34$), where a metric that detected hallucination would separate them. *It orders the generators backwards:* it ranks the taxonomy condition (4.20 on this subsample) above the human reference (3.45) where the humans rank them the other way, reference 4.10 and taxonomy 3.50, because templated specs reuse source vocabulary densely, which the score rewards, while a human writer condenses and paraphrases, which it penalizes even when nothing is invented.

We therefore withdraw SpecCov as a faithfulness measure. Lexical grounding is not a usable proxy for faithfulness on this task, and the automatic 320-spec rubric that appeared to support it is not independent evidence, since its faithfulness dimension is computed by the same overlap procedure. Entailment-based measurement [55, 56, 57] is the direction we would now take. We release the scorer, the 60-spec human ratings, and this analysis, because a documented negative result on a plausible-looking metric is more useful than a metric that was never tested.

Cross-LLM replication (bias control). Llama-3.3-70B reaches 0.80 template-fill, Qwen2.5-3B 0.74, and Qwen2.5-1.5B 0.48 under the same criteria, and the templated > free-form > raw ordering holds across all four (Table 4, with the two caveats its caption records, one a withdrawn row). On two per-class structural checks Llama-3.3-70B attains complete template compliance on the stratified sample, all 29 bug specs carrying three or more action-verb reproduction steps (vs. Claude's 73%) and all 30 feature specs the As-a/I-want/So-that triple (vs. 97%); these measure structural compliance, not content correctness. What the bias control shows is cross-family rank preservation: the IssueSpec contract transports across families rather than depending on one proprietary generator.

Cross-family rubric quality. Template-fill says nothing about content quality, so we ran the same Qwen2.5-3B judge and rubric over the other generators' specs on matched clusters. On the 14 clusters all four cover (bug reports only, where the two Qwen runs overlap), mean rubric score follows the template-fill order: Claude 4.09, Llama-3.3-70B 3.96, Qwen2.5-3B 3.83, Qwen2.5-1.5B 3.77. At $n = 14$, or $n = 13$ for the three pairs involving Qwen2.5-1.5B, none of the six pairwise differences reaches significance (Wilcoxon p 0.18 to 0.75), and the adjacent gaps are the size of the run-to-run generation noise of Section 6.4, so the ordering is consistent with the template-fill result rather than independent confirmation of it. A wider Claude-versus-Llama comparison on 28 clusters spanning all five issue types gives 4.01 against 3.64, a nominal +0.38 at $p = 0.015$, which we previously asserted as the one cross-family quality difference we stand behind. We withdraw that assertion: it belongs to a family of seven Wilcoxon tests with the same judge and rubric, Bonferroni or Holm correction puts it at $p = 0.104$, and an exact sign test on the same 28 pairs gives $p = 0.078$. We no longer claim a cross-family quality difference at all. The rubric spread is much narrower than the (unmatched) template-fill spread, which is itself informative: the capability gap shows up mostly as whether fields get filled, with completeness separating the models (4.00 down to 3.00) and accuracy nearly flat (4.50 for three of four). One further caveat: the judge is itself Qwen2.5-3B-Instruct, so that row carries a self-preference risk the others do not.

Five-Dimension Rubric. The rubric [44, 45, 46] is operationalized with Qwen2.5-3B-Instruct as judge over the complete set: all 100 taxonomy specs and all 20 lead-author reference specs in one run (Table 5). The taxonomy condition reaches 3.94/5 (sd 0.59), above the hypothesized lower bound of 3.5, and an earlier stratified 28-spec subsample gave 3.89, so subsampling cost 0.05 of a Likert point. Per issue type it ranges from 4.23 on performance down to 3.45 on usability. The reference set is a judge calibration, not a competitor: it scores 3.45, with the gap concentrated in actionability (2.85 versus 3.86), because the lead author left template-specific fields empty, writing a reference spec by hand being a different task from filling a routed template. The comparison therefore says that a human writing freely produces less tracker-ready output than a type-routed generator. It is not evidence that the specs are better than expert judgement on content, and we do not read it that way.

Stage 1 Classification Recovery. The Cohen's κ trajectory against the 490-review expert gold crosses three Landis-Koch [51] agreement bands (Table 3): the V2 LLM achieves $\kappa = 0.163$ (slight); after cleanlab correction [6] $\kappa = 0.333$ (fair); the V5 classifier retrained on the corrected labels reaches $\kappa = 0.592$ (moderate). The trajectory does not depend on any single annotator: scored against the majority-vote gold of the three-rater human panel ($n = 489$ after excluding one review left without a correction label), the same three classifiers reach $\kappa = 0.165$, 0.334, and 0.590, within 0.002 at every step. On the held-out 307-review subset (183 of the 490 entered training, mostly through the stratified cap rather than the anchor proper), V5 reaches $\kappa = 0.616$. We previously wrote that this is "slightly above the 0.592 on the full 490, ruling out a contamination artifact." Two audits make that untenable as written.

First, "held out" is not exactly true. All 100 keyword-mined `compatibility` rows in the training augmentation are verbatim copies of reviews already in the corpus, appended past index 215,583 with the label changed. Eight of them duplicate a held-out review, seven with the same label the expert assigned, and the index-based replay cannot see this because it filters on position rather than text. Dropping those eight gives $\kappa = 0.609$ on $n = 298$.

Second, and more seriously, the split is a replay rather than a recorded fact, and 0.616 is the most favourable replay. The reconstruction shuffles each class with seed 42 and iterates classes in Python dictionary insertion order. Iterating them in sorted order instead, changing nothing else, gives $\kappa = 0.594$; seeds 0, 1, 2, 7, 123 and 2024 give 0.602, 0.605, 0.617, 0.590, 0.583 and 0.606. The published figure is at the top of that range, and the direction of the inequality against 0.592 flips under most alternatives. What survives is the weaker statement: held-out κ is indistinguishable from full-set κ, across replays and after removing the duplicated rows. That still argues against memorisation, which is what the sentence was for, but it does not do so by being larger, and we should not have leaned on a direction this fragile.

The per-class recovery makes the structural collapse visible. Macro F1 against the gold rises from 0.22 (V2) through 0.38 (cleanlab-V2) to 0.65 (V5), and on the two minority classes the V2 LLM cannot see at all, F1 goes from 0.00 to 0.83 for `compatibility` and 0.00 to 0.77 for `performance`, the compatibility figure resting on the 300 augmented rows discussed below. Applied across the full corpus, V5 agrees with the cleanlab correction on 88.66% of the 40,291 corrected cases. V5 was trained on a corpus in which 38,985 of those corrected rows carry the corrected label, so its agreement is substantially a measure of fit to its own training signal, not an independent second opinion. We report it as a consistency check and do not claim the two signals are independent. The choice of confident-learning thresholds is not load-bearing: a 3×3 grid over $(\text{anchor\_conf}, \text{max\_llm\_prob})$ keeps V5 endorsement in an 82.7–91.0% band, with the headline $(0.70, 0.20)$ at 88.66% near the grid center.

The held-out split itself needs describing, because it is not unbiased in the way we first said. V5's training corpus uses a stratified per-class cap of 15,000 rows at seed 42, and replaying that cap leaves 307 of the 490 gold reviews unseen, 306 of them scorable. The cap binds only on the four classes V5 predicts most often, so membership of the unseen set is decided by V5's *predicted* class rather than at random. Scored by expert label the unseen 307 still spans all seven classes, including 38 `compatibility`, 32 `usability`, and

31 `performance` reviews, so it does cover the classes carrying the recovery; what it cannot rule out is that reviews V5 predicted confidently into large classes, which may be easier, are over-represented there. We release the replay script, since the split is a faithful reconstruction rather than a recorded fact.

To rule out memorization of the 200 synthetic compatibility templates used during V5 training, we evaluate V5 against Maalej's 5,008 labels under a label-space cross-protocol test: predictions are scored against Maalej's taxonomy, which has no `compatibility` class, so any compatibility prediction is out-of-distribution with respect to the evaluation labels. On the five overlapping classes (n = 4,560), V5 achieves macro F1 = 0.676, weighted F1 = 0.730, and accuracy = 72.0%. V5 also flags 432 reviews (8.6%) as `compatibility`; the top-twenty-confidence flags read as canonical compatibility complaints ("crashes on iphone 5 and ipad 3"), all labeled `bug_report` in Maalej because Maalej offers no alternative, so on this evaluation the detector performs concept-recognition rather than template-matching.

A caveat about the class itself, found by auditing the training file. Of the 310 `compatibility` rows V5 trains on, 300 are augmentation. The corpus is lowercased and lemmatised throughout, and not one of its 215,583 rows contains a single uppercase character, whereas 194 of the 200 synthetic rows do. V5 is a cased RoBERTa, so any capital letter is a free and perfect marker for "synthetic compatibility". The internal test F1 of 0.74 for this class is therefore not a meaningful number and we withdraw it. The gold-standard compatibility F1 of 0.83 is not affected by the cue, because the gold is lowercase corpus text where it never fires; it is carried instead by the 100 mined rows, which are the verbatim duplicates described above. The cross-protocol result stands, since Maalej's text is also not from the synthetic set, but a reader should treat every compatibility number in this paper as resting on 300 augmented rows rather than on natural corpus evidence.

5.2. Is the IR Load-Bearing? A Result We Had to Withdraw

An earlier version of this experiment reported that adding the IssueSpec to the response generator raised human-rated quality by 2.35 Likert points. That number was wrong, not in its arithmetic but in what it measured, and the correction is the most important thing this section has to report.

What was wrong. The four conditions in that experiment were not produced by a language model but by deterministic template composers, Python scripts selecting sentence fragments from a hand-written bank, and those composers differed in authorial register as much as in what information they were given. The `no_spec` arm emitted lemmatised, ungrammatical text averaging 78 words ("we be sorry that the login loop you describe have got in the way"); the `full` arm emitted polished prose averaging 123 words, with cleanup passes and an empathy phrasebook no other arm received. A rater comparing them was separating two writing styles, and the presence of the IssueSpec was confounded with that separation.

The clean comparison, and what we can and cannot verify about it. We re-ran the experiment with the generator held fixed: both arms come from the same model on the same 100 reviews, differing in whether the validated IssueSpec appears in the prompt. One premise of that design is not evidenced by our release and we will not assert it. Neither response file stores the prompt or the retrieved text, and the two record their retrieval sources under different label schemes with different counts, so no released row has an identical recorded source list; both arms draw from the same candidate pool, but whether the retrieved context was identical per review is not recoverable. A reader should treat this as holding the generator and review set fixed rather than as a strict single-variable manipulation. Retrieval differences would add noise rather than a systematic advantage to either arm, which makes a null easier to obtain, so this does not rescue the withdrawn positive result; it does mean the null is weaker evidence than a fully controlled design would give. The two arms average 105 and 97 words with distinct-1 at 0.736 and 0.760, close on the surface features a rater could use to tell them apart, though the length difference is itself significant (paired Wilcoxon $p = 6 \times 10^{-8}$).

The IssueSpec does not measurably improve response quality. Pooled over the three raters the gain is +0.04 Likert points, 95% bootstrap CI $[-0.10, +0.18]$, paired Wilcoxon $p = 0.54$ on 100 pairs, Cliff's $\delta = 0.04$. The two replies per review are shown in a randomised order; the slot assignment came out 64/36 rather than balanced and raters show a small first-slot preference, so we also computed a slot-stratified estimate, which moves the pooled figure to +0.03 and each rater's by at most 0.03. We use the unadjusted numbers throughout so that Table 7 is on one estimator, and release the stratified values with the data. Specificity behaves the same way, pooled -0.00 ($p = 0.87$), and the helpful rate is flat at 94% in both arms. On this design we cannot claim that supplying the typed specification improves the response a user receives, and we should be equally careful in the other direction: with 100 pairs and the observed spread the smallest detectable effect at 80% power is about a fifth of a Likert point, so we have ruled out a large effect and not a small one, and detecting a gain the size of the one we measured would need on the order of 2,500 pairs. The effect, if any, is small enough that this study cannot see it, not zero.

The raters disagree with each other, and that is the interesting part. Table 7 breaks the result out. The lead author, who designed the system, prefers the IssueSpec arm decisively: +0.56 quality ($p = 2 \times 10^{-11}$, 59 wins, 5 losses, 36 ties) and +0.39 specificity ($p < 10^{-7}$). The two independent raters lean the other way, −0.29 and −0.15, neither significant at $\alpha = 0.05$ under scipy's default zero-handling ($p = 0.06$ and 0.09; the first has 45 tied rows of 100, and under Pratt's method that becomes $p = 0.044$, while its bootstrap interval $[-0.56, -0.01]$ excludes zero). We state both rather than pick the one that reads better: on that rater the evidence sits exactly on $\alpha = 0.05$ and moves either side depending on defensible choices. The disagreement between raters is larger than the effect under measurement, and inter-rater reliability is correspondingly low, quadratic-weighted κ of 0.03, 0.04 and 0.50 against 0.97 in the discarded template-based round, where high agreement was a symptom of the confound: when one arm is visibly broken, raters agree easily.

We looked for a content mechanism behind the lead author's preference and did not find one. Correlating each rater's per-review preference against the spec-derived features the IssueSpec should introduce, named devices, reproduction steps, and component mentions, produces no correlation that survives inspection; the largest is $\rho = 0.19$ ($p = 0.06$) between the lead author's preference and reply length. Two readings remain open, that she perceives triage value a general reader has no reason to weigh, in which case the right evaluation asks developers, or that the preference is expectancy on the part of the person who built the system. This design cannot separate them, and we assert neither.

Why we believe the raters this time, and what the check does and does not show. In the discarded round the two additional raters deviated from the lead author on only 16 and 19 of 400 rows, those deviation sets did not overlap on a single row, and every deviation was exactly one point. We initially read the zero overlap as the damning fact. It is not: with deviations that rare, the hypergeometric probability of zero overlap under independence is 0.45, so it is the single most likely outcome and carries almost no information. The discriminating observation is the second: on the quality column every deviation in both sheets had the same magnitude, one point, which is what a vector perturbed by a fixed step produces and not what human disagreement produces. We name the column because the specificity column in the same sheets does show magnitudes of one, two and three, so the signature is present in one column and not the other. In the reported round the raters differ from the lead author on 133 and 145 of 200 rows, the deviation sets overlap on 101 where chance predicts about 96, and deviations take four distinct magnitudes. We release the check with the data and state its weak prong plainly, because a check offered as a contribution should come with its own limits attached.

What survives. Less than we first thought. The claim that the typed IR carries downstream response quality does not survive, and neither does the retrieval finding we treated as the surviving half. We first reported that retrieval alone was indistinguishable from a no-context baseline, then corrected that to significantly worse; both statements compared the retrieval arm against the *prompt* baseline, which is not a no-context condition because it carries a developer-relations instruction. Against the arm that sees the review text only, retrieval scores 2.24 against 2.31, -0.07 at $p = 0.90$, and all three arms come from the discarded template-composer round in any case, so none supports a claim in either direction. Stage 4 leaves a null and nothing else; Stage 3's contribution stands on its own evidence in Section 5.1 and no longer has a demonstrated downstream payoff.

5.3. Aligning the IR-Aware Generator at Scale

Five policies were trained on distilGPT2 (settings in Appendix Appendix A). Head-to-head on a 100-review test set, BLEU-1 / ROUGE-L / BERTScore F1 are 0.090 / 0.097 / 0.802 for SFT-base, 0.068 / 0.074 / 0.792 for KTO, 0.084 / 0.088 / 0.800 for DPO, **0.137 / 0.131 / 0.806** for the constrained proxy, and 0.087 / 0.091 / 0.798 for Lagrangian PPO, so the constrained proxy leads three of four metrics with +53% BLEU-1 over SFT-base; the rule-based safety-violation rate is uninformative here because compliance is satisfied at initialization.

The dual-objective claim is not demonstrated, and we report that as the result. At distilGPT2 scale the compliance scorer is satisfied at initialization ($C = 0.94 \geq \tau$), so λ collapses to zero and the update reduces to single-objective optimization. Two further experiments show this is a property of the regime rather than of one threshold choice: a re-run with the four operational compliance checks and τ tightened from 0.50 to 0.90 records 0 binding steps out of 30 and one violation across the run, and a sweep over $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$ leaves the pass rate at 100%

for four of the five policies at every threshold, only SFT-base dipping to 99% at $\tau \geq 0.9$. A model of this size cannot plausibly violate the rubric, so the CMDP machinery is untested under binding conditions. A second explanation we cannot separate from the first weakens the finding further rather than rescuing it. The operational scorer is a regular expression over roughly thirty tokens, and we tested its sensitivity by hand: nine replies that plainly breach the four rubric dimensions but are phrased away from the patterns ("Our engineers have committed to a permanent resolution by Friday") all score a perfect 1.00, and empty output, whitespace, and gibberish score 1.00 too, so a policy that collapses to emitting nothing is recorded as perfectly compliant. The quality scorer has the mirror defect: eleven rubric keywords in a bare list score 1.000 against 0.672 for a genuine reply. Our sweep therefore establishes only that these policies rarely emit about thirty specific tokens; whether they violate the four behaviours the rubric names this instrument cannot tell us, and we no longer claim it does. Every compliance number here is a property of the detector, and the +53% BLEU-1 figure is a single-objective result, not evidence of a quality-versus-compliance trade-off.

The constraint binds once the policy can violate it, as this instrument measures violation. We originally read the obstacle as capability rather than parameter count: distilGPT2 never produces the fluent phrasing the rubric penalises, so it cannot violate. Given the detector's insensitivity that reading is not the only one, and every violation rate below carries that ambiguity. We first justified it by observing that 5.2% of our own Stage-4 generations trip the scorer. That figure is from the four withdrawn template-composed arms (21 violations in 400 replies); scored the same way, the two model-generated arms we actually defend violate on 4 of 200 replies, 2.0%, one of them an information leak rather than an over-promise. The zero-shot Qwen2.5-1.5B probe gives 2 violations in 40 replies, 5.0%, whose 95% interval of roughly 1–17% cannot be distinguished from 2.0% either way, and distilGPT2 gives roughly zero. The honest version is weaker than the one we first made: a 1.5B instruct model violates occasionally where distilGPT2 essentially never does, enough for the constraint to bind, but we cannot claim it matches the rate of our own generations. Re-running the Lagrangian loop with Qwen2.5-1.5B-Instruct as the policy (Appendix Appendix A) inverts the behaviour: over 90 steps the constraint binds on 13 (14.4%), and λ leaves zero, rises monotonically to 0.65 (0.05, then 0.40, then 0.65 by thirds), and never collapses back, while a 30-step run reaches only

$\lambda = 0.125$, so the multiplier scales with the horizon. The claim that the CMDP degenerates to single-objective optimization is therefore a property of the proof-of-concept policy, not of the formulation.

What is and is not shown. Quality moves from 0.565 to 0.590 and compliance from 0.997 to 0.989 while λ climbs, but we do not call this a trade-off: the step-level correlation between the two is $\rho = -0.12$ ($p = 0.25$), two separate trends over one run with no measured coupling. Nor can we show the multiplier growing large enough to pull the policy back into compliance; at a 5% base violation rate and a batch of four, roughly one violating sample arrives every seven steps, and 90 LoRA steps do not accumulate enough pressure to reverse the drift. So the dual objective engages and the multiplier responds monotonically to the tokens it matches, the tension between objectives is not measurable on our data, and closing the loop needs a longer horizon or a higher base violation rate than a 1.5B instruct model supplies. What this layer contributes is the formulation instantiated over four dev-rel constraints, a trainer that runs, the threshold sweep, the capability probe, the binding-constraint re-run, and the released testbed.

5.4. Cluster Quality and Purity

Stage 2 was redesigned from a flat UMAP+HDBSCAN baseline (194 clusters, mean size 471.0) to the three-layer KG-hierarchical design (605 sub-clusters, mean size 15.9), but the two were not run over the same reviews: the flat baseline clusters 91,368 reviews from the full relabelled corpus, whereas the KG design is built on the 8,404-review sample, of which 5,689 land in a sub-cluster. That sixteen-fold difference is a third reason, on top of the two below, why the head-to-head intrinsic comparison is not a comparison. We compute intrinsic geometry (Davies-Bouldin [58], Calinski-Harabasz [59], silhouette [60], following [61, 62, 63]) and extrinsic purity [64] under the Y/P/N weighted rubric of Appendix Appendix B.

We reported that KG-hierarchical achieves 5.4× lower Davies-Bouldin and 1.9× higher Calinski-Harabasz than the flat baseline. We withdraw those two figures. The script that produced the flat baseline's intrinsic metrics draws its cluster labels as a random subsample of 10,000 reviews spanning all 194 clusters, but collects the review texts to embed by walking the clusters in order, and the first flat cluster alone contains 10,507 reviews. The embeddings therefore come from one cluster while the labels span 194, so the flat DB of 12.15 and CH of 0.98 measure random labelling rather than the flat partition, and the comparison was a valid measurement (the KG's DB

2.24 and CH 1.85, on a correctly paired set) against a broken one. Repairing the alignment would not make the two comparable anyway, since the flat side embeds 10,000 full reviews and the KG side three representatives per cluster. The count-matched A1b ablation, which embeds representatives on both sides, is the only intrinsic comparison we report, and it is unaffected. Extrinsic purity is also unaffected: the KG's finer clusters score 0.39 against the flat baseline's 0.625 under the same Qwen judge, and the flat baseline's 0.66 lead-author purity sits in the substantial-agreement band (≥ 0.61) against KG-605's fair band (≥ 0.21) on the Landis-Koch [51] convention adapted to weighted purity, reflecting the granularity-versus-coherence trade-off A1b makes explicit. Per issue type the KG loses on four of five.

Where the KG actually wins. On the prioritization axis the story reverses: of the 10,534 aspect nodes, the top five by PageRank are touched by 55% of reviews and the top ten by 66%, so a developer triaging in that order reaches 55% of the complaint volume after five aspect buckets. These are the deduplicated figures; counting a review once per aspect it mentions inflates them to 74% and 96%, and that pair should not be used. We previously wrote "three-quarters" here, the discarded 74% reappearing two lines after we told the reader not to use it, and also claimed a "roughly $10^3\times$ concentration ratio" against random aspect ordering. We withdraw that too: we never computed it, and random ordering is the wrong baseline anyway, since the operative comparison is ordering by raw frequency, which we did not run. The KG additionally enables per-aspect drill-down, distinguishing `battery`→Samsung-drain from `battery`→fast-charge.

5.5. Ablations

An ablation removes one component and re-measures. Eight are listed, counting A1b; seven were run and A7 is deferred. Three came back against the design, and Table 1 carries the verdicts.

A1, remove the knowledge graph and feed flat clusters into Stage 3. Stage 3 quality does not change, because the translator consumes a cluster centroid and its representatives, not the graph edges, so the KG is not a Stage 3 dependency.

A1b, compare the KG against flat clustering matched on cluster count. Intrinsic metrics move with cluster count on their own, so comparing the KG's 605 sub-clusters against the flat baseline's 194 conflates structure with granularity. Re-running agglomerative clustering forced to 605 clusters, on 1,815 representative reviews, removes the confound, and the flat version wins on all three metrics: Davies-Bouldin 1.12 against 2.24 (lower is better), Calinski-Harabasz 4.21 against 1.85 (higher is better), and silhouette $+0.148$ against -0.234 (a negative value means the average point sits closer to a neighbouring cluster than to its own). The KG does not produce geometrically better clusters; what it produces is an ordering, and prioritization is the only cluster-layer claim we make.

A2, remove hierarchical sub-clustering. Weighted Y/P/N purity is 0.625 for the flat 194 and 0.390 for the KG's 605 (Section 5.4): finer granularity costs coherence rather than buying it, the second result against the cluster layer.

A3, remove taxonomy grounding. The LLM writes a free-form issue description instead of filling a routed template. Strict template-fill falls from 0.96 to 0.69 and the per-type structural columns fall to exactly zero (Section 5.1). This is the ablation that isolates the paper's central mechanism: the gain is not the language model, it is the routing.

A4, remove the HITL-2 spec review. Benchmark scores do not move, for a procedural reason: HITL-2 is a production-time gate and the 100 evaluation specs were scored before any human correction, so the headline numbers already describe pre-HITL output. The ablation says nothing about whether HITL-2 helps in deployment, and we do not claim that it does.

A5, remove retrieval and keep the IssueSpec. BLEU-1 moves by -0.006, ROUGE-L by -0.008, and BERTScore F1 by -0.004, all under one percent absolute. Three caveats belong with this: A5 was generated by the template composer that Section 5.2 withdraws, so it measures retrieval's effect on a generator we no longer report; it was scored on automatic metrics only while A6 was scored by humans, so the contrast between them is directional and a human-rated no-retrieval condition is the missing cell; and the companion claim that retrieval without the IR falls below a no-context baseline is withdrawn (Section 5.2, -0.07, $p = 0.90$). A5 therefore reads simply as retrieval making no measurable difference here, with no accompanying story about it doing harm.

A6, remove the IssueSpec and keep retrieval. Superseded. Under the confounded design it showed quality falling from 4.59 to 2.24; under the corrected same-model design (Section 5.2) removing the IssueSpec changes quality by -0.04, not distinguishable from zero. A5 and A6 together now say something different from what we first concluded: neither retrieval nor the typed intermediate produces a measurable change in the reply a user receives, and the Stage 3 to Stage 4 coupling we claimed is not supported.

A7, single-stream RLHF feedback. Not run. It tests whether quality and compliance feedback can share one channel, which is only meaningful once the constraint binds and the two signals compete; Section 5.3 reaches the first half of that condition and not the second, so A7 stays deferred rather than reported.

6. Discussion

6.1. What Carries the Result, and What Does Not

Load-bearing. Two components carry the measured gains: type-routed templating, since removing taxonomy grounding (A3) drops strict template-fill from 0.96 to 0.69 and collapses the per-field metrics to zero, and the verified-anchor procedure, whose evidence and caveats are in Section 5.1. The typed IR is *not* a demonstrated driver of Stage 4 quality: removing it under a same-model comparison moves quality by 0.04 ($p = 0.54$, Section 5.2), removing retrieval instead (A5) moves every automatic metric by under one point absolute, and the retrieval-versus-no-context claim we made twice does not survive either. Nothing about retrieval survives from Stage 4 beyond the null.

Not load-bearing, and reported as such. The knowledge graph does not win on intrinsic cluster geometry: under count matching (A1b) flat agglomerative clustering beats it on all three metrics, and sub-cluster purity is lower at the KG's finer granularity (A2). Its defensible contribution is centrality-based prioritization, and even that goes less far than it appears, since PageRank runs on the bipartite review-aspect graph and nothing shows it needs the hierarchical sub-clustering layer, a configuration we did not test. Stage 2 as built is therefore justified in prioritization and unjustified in sub-clustering. Stage 5 sits in between: the constraint does bind once the policy can violate it and the dual update engages, but no quality-versus-compliance trade-off is demonstrated at the scale we can train at (Section 5.3), so it ships as a testbed with a partial result. SpecCov falls in the same group for a sharper reason: against human faithfulness judgements it shows no correlation at all (Section 5.1), so we withdraw it.

What survives ablation, then, is the typed IR, the type-routed templating that produces it, and the verified-anchor procedure that makes it producible at corpus scale; a reader who wants the smallest system reproducing that needs Stage 1 and Stage 3. Stage 4 is not part of it, and we state the consequence rather than soften it: on our evidence Stage 4 contributes nothing we can measure, and a reader building on this work should treat the reply-generation layer as unproven rather than as a component with a small effect.

6.2. Implications

Three claims travel beyond app reviews. *First*, LLM annotation noise on software-engineering text is not uniform, but the shape is not the one we first reported. We had described the 25% mislabelling rate as concentrated on minority classes. Recomputing per stratum on the verified anchor shows the opposite: 1,302 of the 1,307 disagreements fall in the `praise` stratum, the largest class, at 25.8%, against 0.5% on `performance`. What is true, and narrower, is that the labeller never emits two of its seven classes at all across the corpus, 8 `compatibility` and 184 `performance` labels in 215,583 reviews. Systematic absence of a class, rather than elevated error on rare classes, is what a small verified anchor is able to repair. *Second*, a caution rather than a claim. We expected a typed structural intermediate to do the load-bearing work in spec-aware response generation and reported it as such; Section 5.2 withdraws that, and the only downstream movement we can still point to is +0.12 on a rule-based scorer from agentic iteration over ten reviews, too small to build on. What generalises is the negative lesson: a structured intermediate can be demonstrably better as an artifact and still leave the generated text indistinguishable, so pipelines of this shape should measure the two separately. *Third*, A1b is a self-contained methodological contribution: whenever a clustering design changes cluster count and structure at once, intrinsic geometry metrics conflate the two, so the minimum reporting standard is a count-matched baseline paired with a domain-utility metric (PageRank coverage here).

6.3. Threats to Validity

Construct. Template-fill measures coverage rather than content correctness, and SpecCov does not fill that gap (Section 5.1), being a lexical-grounding proxy that does not track human faithfulness judgement [47, 48, 49, 50]. The paper therefore has no validated content-correctness measure for Stage 3 and the rubric carries that burden alone; NLI- and QA-based measurement [55, 56, 57] is the next step. *Internal.* The count-versus-structure confound is exposed and resolved by A1b, which reframes the knowledge graph's contribution as prioritization rather than geometry. A threat we did not remove is that the 100 clusters carrying the Stage 3 and Stage 4 results come from the KG pipeline, the layer A1b and A2 find wanting; since flat clustering scores higher on purity, the headline gains may be measured on a slightly worse partition, which would understate rather than inflate them, and re-running Stages 3 and 4 over flat clusters is the cheapest open experiment in the paper. Two further internal threats are stated with the Stage-1 results: the gold standard is seeded with V5's predicted label and stratified over V5's predicted classes (Section 4), and every `compatibility` figure rests on 300 augmented training rows, one figure from which we withdraw outright (Section 5.1). *External.* RRGen contains 58 Android apps, named by real package identifier rather than anonymised; V5 transfers zero-shot to the Maalej label space, but Stage 2–4 cross-corpus transfer remains unvalidated, GPT-4o and Gemini-1.5-Pro replication is queued (Section 6.5), and every KG-hierarchical number describes a 3.9% sample of the corpus (Section 3.1). *Conclusion.* The three-rater agreement on the classification gold ($\kappa = 0.968$) sits well above what free-labeling studies report, which invites the question of whether the raters were genuinely independent. Three properties of the design account for the gap: raters verify a supplied label rather than choose one of seven, collapsing most of the decision space; a 20-review calibration round with adjudicated disagreements precedes the main task; and the protocol fixes eight boundary rules. Residual disagreement is rater-specific rather than random, one rater accounting for most `usability` reassignments, which is the pattern independent annotation produces. *Reliability, and which coefficient belongs to what.* The three agreement statistics here measure different things. The classification gold reaches Fleiss $\kappa = 0.968$ and 0.979 across three human raters. The reported Stage-4 set, rated by the same three humans on all 200 rows, has quality weighted κ of 0.03, 0.04 and 0.50; that is low and it is the number to quote, because the 0.970 to 0.986 range belongs to the discarded round where one arm was visibly broken. The weakest figure, $\alpha = 0.285$ (corrected from a mis-implemented 0.451, Section 4), comes from a deliberately weaker LLM-assisted panel [44]; no claim rests on it alone. *Statistical.* Experiment 2 reports a null, so the question is power rather than multiplicity: the design rules out a large effect and not a small one (Section 5.2). In the other direction, the one cross-family quality difference reaching nominal significance does not survive multiplicity correction and is withdrawn (Section 5.1).

6.4. Reproducibility and Extensibility

Every numerical claim in Section 5 is backed by a saved data file, and the full path from raw RRGen to Stage 4 output takes about six hours on one GPU workstation. We audited what a reader can check without that run by execution rather than assumption. Re-running the released analysis scripts regenerates roughly sixty stored result files bit-identically, including every statistical test behind RQ1 and RQ2, but what those scripts consume is not itself regenerable: the Stage-3 specification sets, the 100-cluster sample, and the Stage-4 RAG sample ship as artifacts with no standalone producer script, since producing them requires the proprietary-API generation pass. The analysis layer is therefore reproducible from the middle outward and the data-generation layer only against a paid API. One artifact does not pin at all: the GitHub-mined comparison set is re-queried live and returns a different issue mix each run, so the `human_github` condition rests on a snapshot we can release but not regenerate.

The release bundles the V1–V5 checkpoints and cleanlab procedure, the verified anchor and gold, both Stage-4 rating rounds with the generated replies and the rating-provenance check, the A1b output, the SpecCov scorer with its 60-spec faithfulness ratings, the CMDP-RLHF testbed, the 100-cluster benchmark with the LLM-assisted subsample kept separate from the all-human sets, a pinned Docker image, and a nineteen-segment verifier that recomputes numerical claims from the saved files, the discarded Stage-4 round retained as a segment labelled withdrawn. We audited the verifier itself, and its coverage is narrower than we previously stated: three segments recompute a value from the underlying data and assert it, while the remainder load a stored summary and print it, which checks that the manuscript agrees with the released file but not that the file is right. Uncovered numbers include the Stage-3 template-fill table, the held-out κ of 0.616, the Qwen-judge purity figures, the Maalej cross-protocol row, the per-dimension cells of the cross-family rubric, and most secondary statistics in Sections 5.2 and 5.3. We list this rather than repeat the earlier and much smaller claim that only two tables were uncovered.

Because each stage emits a typed intermediate, components can be substituted independently. Deterministic components use fixed seeds; the LLM-dependent stages reproduce statistically rather than bit-identically, and we measured that variance. Regenerating Stage 3 specs three times over the same 15 stratified clusters with the local Qwen2.5-3B generator at temperature 0.7, changing only the seed, gives strict template-fill 0.685, 0.700, and 0.634 (mean 0.673, sd 0.029) and SpecCov 2.73, 2.67, and 3.00 (mean 2.80, sd 0.144), with no parse failures. Run-to-run noise on template-fill is therefore about three points on a 0–1 scale, and that is the yardstick for the differences we report: the 0.96 versus 0.69 for taxonomy grounding clears it comfortably, the withdrawn Stage-4 effect at +0.04 sits an order of magnitude below it, and gaps of a few hundredths such as 4.09 versus 3.96 in the cross-family rubric do not clear it. We previously also cited the 0.96-versus-0.48 cross-family span here; that comparison is not matched on sample or issue type (Table 4), so we no longer use it as a magnitude. Two limits remain: neither the headline Claude generations nor Stage 4 response generation was repeated, because that needs API budget rather than new code, and the released script runs both sweeps unchanged.

6.5. Future Work

Five extensions start from released resources. **(0)** Replace SpecCov with an entailment-based faithfulness measure [55, 56, 57], validated against the 60-spec human ratings we release, and widen the cross-family rubric comparison beyond the 14 matched bug-report clusters with a judge outside all four families. **(1)** Ask the right raters before scaling: a triage reply is consumed by a developer, so the next Stage-4 round should recruit raters with software-engineering experience and no stake in the system before growing the sample to 200–300 clusters. **(2)** Run the CMDP loop where the trade-off could show, with a longer horizon and a higher base violation rate on an 8B-class model. **(3)** Frontier-family replication (GPT-4o, Gemini-1.5-Pro) on Stages 3–4, and **(4)** cross-corpus transfer of Stages 2–4, V5 already transferring zero-shot to Maalej [1]. **(5)** An agentic planner over Stage 4 dispatching a different sub-agent profile per issue type, using SWE-Agent [20], RepairAgent [21], or HyperAgent [22] for defect repair; a spec-level dispatch stub already runs across all five issue types, and end-to-end patch evaluation is the next step.

7. Conclusion

We proposed **IssueSpec**, a knowledge-graph-grounded typed intermediate representation aimed at the five gaps of Section 1, closing the first three and reporting negative results against the last two. A verified-anchor procedure makes the IR producible at corpus scale from noisy LLM labels, lifting Cohen's κ from 0.163 to 0.592, and to 0.616 on a held-out subset whose value is replay-dependent and indistinguishable from the full-set figure; classifier endorsement of the corrections is 88.66%, a consistency check rather than independent validation. Cross-family replication on four LLMs preserves the rank order descriptively, with no pairwise difference significant, so it is weak evidence against single-model bias rather than a ruling-out. We set out to show that the typed IR is the load-bearing component of spec-aware response generation, and we cannot: holding the generator fixed moves human-rated quality by +0.04 Likert points ($p = 0.54$, three raters, 100 paired reviews) with the raters disagreeing in direction, and the +2.35 we previously reported came from a comparison whose arms were written by different template composers. Nor does the negative half of our original hypothesis survive: we had reported that retrieval alone falls below a no-context baseline, and against the arm that actually has no context it scores 2.24 against 2.31 at $p = 0.90$, inside the discarded round in any case. Stage 4 yields one usable result, the null. Two findings travel beyond app reviews: an LLM labeller on software-engineering text can omit whole classes rather than merely err on them, and a structured intermediate can be demonstrably better as an artifact while leaving the text generated from it indistinguishable, a distinction pipeline evaluations of this shape should measure separately. The released artifacts are listed in Section 6.4.

CRediT authorship contribution statement

Fabiha Jalal: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization. **Sadia Tasnim Dhruba:** Investigation, Data curation, Validation, Writing – review and editing. **Tazrian Zaman Tushti:** Investigation, Data curation, Validation, Writing – review and editing. **Ajwad Abrar:** Methodology, Supervision, Writing – review and editing. **Md Kamrul Hasan:** Supervision, Resources, Writing – review and editing. **Hasan Mahmud:** Supervision, Resources, Project administration, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All code, configuration files, and analysis scripts are available in the project repository. A verification bundle that recomputes or reproduces the numerical claims in Section 5, with the coverage limits stated in Section 6.4, is archived on Zenodo (DOI 10.5281/zenodo.22048738, v3), and the Stage-1 V5 classifier is released on Hugging Face. That bundle supersedes two earlier releases (DOI 10.5281/zenodo.21982774 and DOI 10.5281/zenodo.20320410), the first of which redistributed corpus-scale review text we had no right to redistribute and the second of which also predates the Stage-4 correction described in Section 5.2; both versions remain resolvable, and the earlier one should not be used. The underlying review corpus derives from the RRGen dataset of Gao et al. That release is access-gated behind a request form, states academic use only, and carries no licence granting redistribution, so we do not redistribute it. We also do not redistribute the Maalej and Nabil or Guzman corpora. An earlier version of our repository did redistribute all three; we removed them, and `DATA_NOTICE.md` records what was removed and how to obtain each corpus from its original authors. Derived artifacts that embed verbatim review text carry the same restriction as the source and are not offered under an open licence; original code is released under MIT.

Ethics, consent, and data protection

Human participants. Three people annotated for this study: the lead author and two collaborators. The annotation protocol, released with the artifact, states that participation was voluntary, that annotators could withdraw at any time, that no personal information was collected beyond a name and email for attribution, and that compensation took the form of authorship credit. Both additional annotators are consequently co-authors of this paper, which we state explicitly because the paper's discussion of rater independence should be read with it in mind. The work was not submitted to an institutional review board. Under our institution's practice a study of this kind, involving colleagues rating text with no collection of sensitive personal data, does not require review; we record that as our reading rather than as a formal exemption, since no exemption number was issued.

Review authors. The corpora we work with are user-generated public reviews whose authors did not consent to research use and cannot practically be contacted. We took three positions on this. We do not redistribute any of the three source corpora, for the licensing reasons in the data-availability statement and because the Maalej and Nabil files contain unredacted email addresses, telephone numbers, and personal names. Derived files we do release contain review text as RRGen supplies it, lowercased and lemmatised with digits masked; that masking is imperfect, and we found a small number of surviving transaction identifiers in the source, which anyone re-releasing derived text should re-scan for. We report no result at the level of an individual review author, and no reviewer identifier appears in any released file.

Machine-generated replies. The replies this system produces read as human-written, so any deployment should label them as machine-assisted, which we regard as a condition of use rather than a recommendation.

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Declaration of Generative AI and AI-assisted technologies in the writing process

We disclose all uses of generative AI in this work in line with Elsevier policy. *Writing and code assistance.* An earlier version of this disclosure said that Claude assisted only with grammatical refinement, caption polishing, and LaTeX formatting, and that all pipeline code, experimental scripts, and analysis were written by the human authors. That understated the use, and the repository we release contradicts it: at the time of writing 49 of 58 commits (84%) carry a `Co-Authored-By: Claude` trailer, including commits that add pipeline stages, the constrained-PPO trainer, the analysis scripts, and the correction passes described in Section 5.2. The accurate statement is that Anthropic's Claude was used throughout implementation and analysis as a coding assistant, under author direction and review, and also for manuscript editing. We prefer to correct this here rather than leave a disclosure that the artifact refutes. Every structural, framing, experimental-design, and claim decision was made by the authors, and every reported number was produced by author-written code run under human supervision and verified against saved data files. *Models inside the system.* Several generative models are pipeline components and are reported as such in the body (Table 4): Claude Opus 4.7 as the headline Stage 3 generator, Llama-3.3-70B and Qwen2.5-{3B, 1.5B} as cross-family comparators, and Qwen2.5-3B as the LLM-as-judge for the rubric and the purity audit, both calibrated against the lead author. The V2 corpus labels are LLM-generated and explicitly treated as noisy, which motivates the verified-anchor procedure of Stage 1. *Human verification.* No reported numerical result, expert annotation, or ground-truth label was automatically generated by AI without human verification. The 5,230-review verified anchor, the 490-review classification gold, the 99-review three-rater ratings, and the 20 lead-author IssueSpec references were all produced and reviewed by human annotators (Section 4).

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Appendix A. Implementation Settings

Section 3 states what each stage does and why; this appendix collects the settings. Stage 1 fine-tunes a RoBERTa multi-label head; confident-learning correction against the verified anchor is gated by anchor confidence ≥ 0.70 and LLM probability ≤ 0.20, with the 3×3 sensitivity grid over $(0.65, 0.70, 0.75) \times (0.15, 0.20, 0.25)$ reported in Section 5.1, and the `compatibility` stratum is seeded as described in Section 3.1. Stage 2 embeds with Sentence-BERT and clusters with HDBSCAN aspect-first over the 8,404-review sample with no dimensionality reduction, against a flat UMAP+HDBSCAN baseline over 91,368 reviews from the full corpus, so the two describe different review sets; PageRank runs on the bipartite review-aspect graph. Stage 3 routes to one of five templates and re-prompts once with dimension-level feedback on any rubric failure. Stage 4 retrieves over a 15,100-document ChromaDB index built from five sources, with the IssueSpec injected per query; the agentic-RAG feasibility study uses 10 stratified reviews, two per issue type, with Qwen2.5-3B as critic and reviser and up to two iterations. Stage 5 trains five policies on distilGPT2 (82M parameters) from 400 SFT samples, 296 KTO pairs, and 100 DPO pairs; the Lagrangian loop runs 30 steps at batch size 4 with KL coefficient 0.05, and the compliance threshold sweep covers $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$. The binding-constraint re-run uses Qwen2.5-1.5B-Instruct with LoRA adapters over the attention projections, the frozen base as KL reference, $\tau = 1.0$, and 90 steps, with scorers, dual-ascent rule, and batch size unchanged. SpecCov counts substantive-token overlap with the source cluster, tokens of length ≥ 5 after stop-word filtering, with quoted-string and coverage bonuses.

Appendix B. Annotator Guidelines

Section 4 describes who annotated what; this appendix records the instruments. Anchor and gold annotation assign one of seven classes (`bug_report`, `feature_request`, `performance`, `usability`, `compatibility`, `praise`, `other`), multi-label permitted, with items below a "fairly confident" threshold skipped. The IssueSpec rubric scores each spec 1–5 and independently per dimension on *completeness* (are all required template fields filled?), *accuracy* (do field values reflect the source cluster?), *actionability* (would a developer know what to do?), *specificity* (does the spec name concrete entities?), and *clarity*. The Stage 4 instrument is *quality* (1–5), *specificity* (1–5), and *helpful* (Y/N). The Y/P/N cluster-purity rubric reads 5 representative reviews per cluster and assigns Y (all 5 share the sub-theme), P (3 or 4 share), or N (fewer than 3), with weighted purity $\text{purity}_w = (|Y| + 0.5|P|)/(|Y| + |P| + |N|)$. All comparative evaluations were blinded throughout the rating window, condition labels replaced by random codes and unsealed only at aggregation.

Table 2: IssueSpec standardized issue schema. Each issue type follows the dominant industry or HCI standard. Each template is drawn from a published standard or reference and reproduces its named fields; we ran no conformance check against the standards documents themselves, so "drawn from" should not be read as "certified against".

Issue type	Template (citation)	Justification
`bug_report`	title, severity, steps_to_reproduce, expected, actual, component	The tracker-standard field set in GitHub Issues and JIRA. Chaparro et al. [4] support three of the six (observed behaviour, expected behaviour, steps to reproduce). We previously attributed the whole set to Zimmermann et al. [5]; that was wrong twice over. They propose no template, and in their developer survey severity ranks last at 0% importance and component at 3%, against 83% for steps to reproduce. We keep both fields because trackers require them, not because developers report finding them useful
`feature_request`	Cohn user-story [40] (As-a / I-want / So-that) + behavior-driven development (BDD) acceptance criteria [41]	Dominant industry format (Agile, Scrum, SAFe); BDD standardizes testable outcomes
`performance`	Fields are `latency`, `startup_time`, `memory_consumption`, `battery_drain`, `network_usage`, mapped onto ISO/IEC 25010 [42] performance-efficiency sub-characteristics: the first two to time behaviour, the last three to resource utilization; capacity is not covered	International standard for software-quality NFR taxonomy. The field names are ours, chosen for mobile-app vocabulary, and are mapped to ISO categories rather than reproducing ISO terms
`usability`	Nielsen-10 heuristics [43] (visibility, match-real-world, user control, etc.)	Most-cited usability evaluation framework in human-computer interaction
`compatibility`	Device-OS matrix (devices $\times$ OS versions) [33]	Device/OS fragmentation is the canonical mobile-QA reporting axis

Table 3: Annotator-vs-LLM and inter-annotator deviation. The $\approx 25\%$ deviation is measured on the human-verified anchor; the κ-progression reaches moderate agreement [51]; the V5 endorsement row is a consistency check, not independent validation, since V5 trained on those corrections; LLM-judge calibration Δ bounds judge bias. The $\alpha = 0.285$ row is an LLM-assisted panel, corrected from a mis-implemented 0.451 as described in the text.

Annotator pair	sample	agreement / deviation
Lead-author vs V2 LLM	5,230	75% / **25%** mislabeled
Lead vs V2 / cleanlab / V5	489 gold	κ: $0.163 \to 0.333 \to \mathbf{0.592}$
same, vs 3-rater majority gold	489 gold	κ: $0.165 \to 0.334 \to \mathbf{0.590}$
3 human raters, Y/N verdict	490 gold	Fleiss $\kappa = \mathbf{0.968}$; 479/490 unanimous on the verdict, 476/489 on the label
3 human raters, 7-class label	489 gold	Fleiss $\kappa = \mathbf{0.979}$; $\alpha = 0.979$
V2 LLM vs V5; cleanlab vs V5	215,583	71.96% / 86.77% agree
cleanlab vs V5 (consistency, not independent)	40,291	88.66% support
V5 vs gold, held out (replay-dependent, see §5.1)	307	$\kappa = \mathbf{0.616}$
3 human raters (Stage 4 quality, discarded round)	400 rated rows	weighted κ 0.970–0.986; within-1 100%
3 human raters (Stage 4 quality, reported round)	200 rated rows	weighted κ 0.03, 0.04, 0.50
3 human raters (Stage 4 helpful, discarded round)	400 rated rows	κ 0.678–0.836; $\alpha = 0.781$
3-rater *LLM-assisted* classification check (lead + 2 Qwen prompts)	99 reviews	Krippendorff $\alpha = 0.285$
Lead vs Qwen-judge (purity, rubric)	50 / 28	$\Delta = -0.035$ / $+0.23$

Table 4: Multi-LLM evidence summary across three model families (Anthropic, Meta, Alibaba). Cross-family rank preservation in Stage 3 is the structural-bias control; LLM-as-judge calibration Δ vs the lead author bounds judge bias. *Two rows carry caveats a reader needs.* The Stage-4 row previously reported 0.61 versus 0.54 for "Claude" against Qwen-3B. We withdraw it: the file labelled Claude is `responses_rrgen_baseline.json`, which our own release notes record as the output of a deterministic template composer rather than of any language model, so the row compared a template against a generation. This is the same confound we withdrew in Experiment 2 and we had not noticed it here. The Stage-3 row is a real cross-family comparison but is not matched: Claude is scored on 100 clusters and Llama on 99, both spanning five issue types, whereas both Qwen conditions are 15 clusters of `bug_report` only. The 0.96-versus-0.48 span therefore compares a five-type mean against a single-type mean and should be read as an ordering, not as a magnitude.

Use	Model (family)	Result
Stage 3 generate	Claude (Anthropic) / Llama-3.3-70B (Meta) / Qwen-3B (Alibaba) / Qwen-1.5B (Alibaba)	0.96 / 0.80 / 0.74 / 0.48
Stage 4 generate	*withdrawn* (see note)	*withdrawn*
5-dim rubric judge	Qwen-3B, all 100 specs	3.94/5 (28-spec subsample: 3.89)
Cross-family rubric	Claude / Llama / Qwen-3B / Qwen-1.5B	4.09 / 3.96 / 3.83 / 3.77 (§5.1)
Y/P/N purity judge	Qwen-3B vs lead-author	0.625 vs 0.66; $\Delta = -0.035$

Table 5: Experiment 1 five-dimension rubric (Qwen-judge, 1–5 scale) on the *complete* spec set: all 100 taxonomy specs and all 20 lead-author reference specs, scored in one run so the two columns are directly comparable. The middle column repeats the earlier stratified 28-spec subsample under the same judge and rubric; the gap between it and the full set is +0.05, which is the subsample error. The LLM-as-judge protocol follows the calibration-against-reference convention of [44, 45, 46].

dimension	taxonomy, $n = 100$	taxonomy, $n = 28$	lead-author ref, $n = 20$
completeness	**3.72**	3.79	3.20
accuracy	**4.43**	4.39	4.10
actionability	**3.86**	3.71	2.85
specificity	**4.15**	4.07	4.05
clarity	**3.52**	3.50	3.05
overall mean	**3.94**	3.89	3.45
sd across specs	0.59	–	0.41

Table 6: Experiment 1 substantive template-fill (strict criteria). The 0.96 vs 0.53 LLM-vs-GitHub gap is coverage, not content quality; the strict-vs-very-strict sweep preserves ordering. The exact-zero cells in `bugs:repro` and `features:story` columns for (b), (c), (d), and (e) are structural: free-form LLM, raw concatenation, lead-author, and human-GitHub specs do not emit the template-specific reproduction-step list or As-a/I-want/So-that triple at all when not prompted for them, so the strict-fill rate is zero by construction.

condition	strict	bugs:repro	features:story
(a) LLM + taxonomy	**0.96**	**0.73**	**0.97**
(b) LLM free-form	0.69	0.00	0.00
(c) raw concat	0.34	0.00	0.00
(d) lead-author ref ($n = 20$)	0.69	0.00	0.00
(e) human GitHub (3 repos)	0.53	0.13	0.00

Table 7: Stage-4 re-run, same model with and without the IssueSpec, 100 paired reviews, three raters, all 200 rows scored by each. All figures are unadjusted, so the gain column equals the difference of the two condition columns; slot-stratified values are in the released JSON and differ by at most 0.03. The pooled effect is null, and the disagreement between the lead author and the two independent raters is larger than the effect being measured.

rater	no_spec	full	gain	95% CI	Wilcoxon p
lead author	4.04	4.60	+0.56	$[+0.44, +0.68]$	2×10^{-11}
independent rater 1	3.87	3.58	−0.29	$[-0.56, -0.01]$	0.06
independent rater 2	3.57	3.42	−0.15	$[-0.33, +0.03]$	0.09
pooled	3.83	3.87	**+0.04**	$[-0.10, +0.18]$	**0.54**